

Foundation (Jalapeno)

- Q1.** If $f(x) = x^2$, which function represents a vertical shift of $f(x)$ by 2 units up?
(A) $f(x) = x^2 + 2$
(B) $f(x) = x^2 - 2$
(C) $f(x) = (x + 2)^2$
(D) $f(x) = (x - 2)^2$
(E) $f(x) = x^2 + 1$
- Q2.** Which transformation reflects $f(x) = x^2$ over the y -axis?
(A) $f(x) = (-x)^2$
(B) $f(x) = x^2 + 1$
(C) $f(x) = -x^2$
(D) $f(x) = (x + 1)^2$
(E) $f(x) = x^2 - 1$
- Q3.** If $f(x) = x^2$, which function represents a horizontal shift of $f(x)$ by 3 units to the left?
(A) $f(x) = (x + 3)^2$
(B) $f(x) = (x - 3)^2$
(C) $f(x) = x^2 + 3$
(D) $f(x) = x^2 - 3$
(E) $f(x) = (x - 2)^2$
- Q4.** Which function represents a stretch of $f(x) = x^2$ by a factor of 2?
(A) $f(x) = 2x^2$
(B) $f(x) = (2x)^2$
(C) $f(x) = x^2 + 2$
(D) $f(x) = x^2 - 2$
(E) $f(x) = \frac{x^2}{2}$
- Q5.** If $f(x) = x^2$, which function represents a compression of $f(x)$ by a factor of $\frac{1}{2}$?
(A) $f(x) = (\frac{x}{2})^2$
(B) $f(x) = \frac{x^2}{2}$
(C) $f(x) = (2x)^2$
(D) $f(x) = x^2 + 2$
(E) $f(x) = x^2 - 2$
- Q6.** Which transformation reflects $f(x) = x^2$ over the x -axis?
(A) $f(x) = -x^2$
(B) $f(x) = x^2 + 1$
(C) $f(x) = -x^2 + 1$
(D) $f(x) = (x + 1)^2$
(E) $f(x) = x^2 - 1$
- Q7.** If $f(x) = x^2$, which function represents a vertical shift of $f(x)$ by 1 unit down?
(A) $f(x) = x^2 + 1$
(B) $f(x) = x^2 - 1$
(C) $f(x) = (x + 1)^2$
(D) $f(x) = (x - 1)^2$
(E) $f(x) = x^2 + 2$
- Q8.** Which function is the parent function of $f(x) = (x - 2)^2 + 1$?
(A) $f(x) = x^2$
(B) $f(x) = x^2 + 1$
(C) $f(x) = x^2 - 1$
(D) $f(x) = (x + 2)^2$
(E) $f(x) = (x - 2)^2$

Open Ended Questions (FRQ)

- Q9.** Graph the function $f(x) = (x - 1)^2 + 2$ and identify the parent function. Provide the equation of the parent function and describe the transformations applied to obtain $f(x)$.
- Q10.** Suppose $g(x)$ is a transformation of $f(x) = x^2$ and has the equation $g(x) = -2(x + 1)^2 - 3$. Describe the transformations applied to $f(x)$ to obtain $g(x)$ and graph both functions on the same coordinate plane.

Chef's Tips

- Emphasize the importance of understanding function transformations.
- Encourage students to use graphing calculators or software to visualize the transformations.
- Remind students to label the axes and provide a key for their graphs.